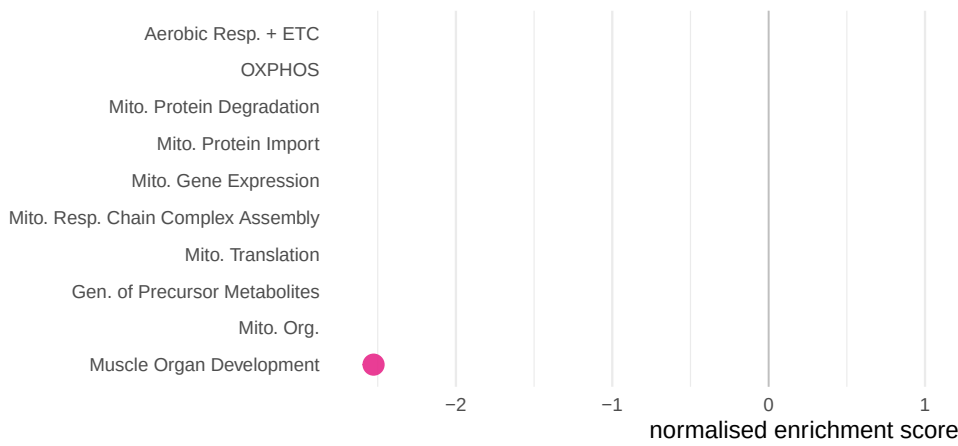


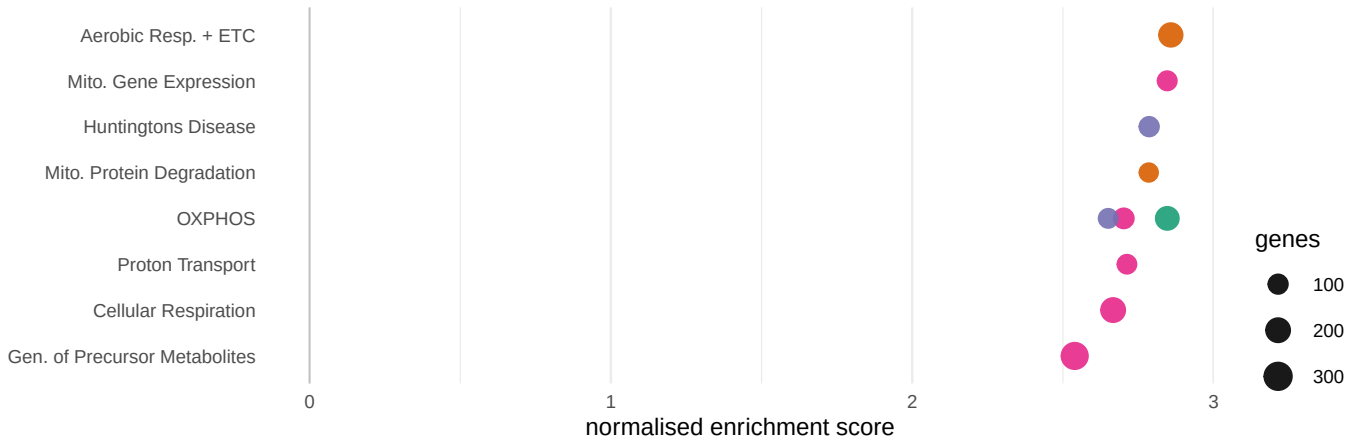
A

## BFR\_Post-Pre, 10 of 88 survivors



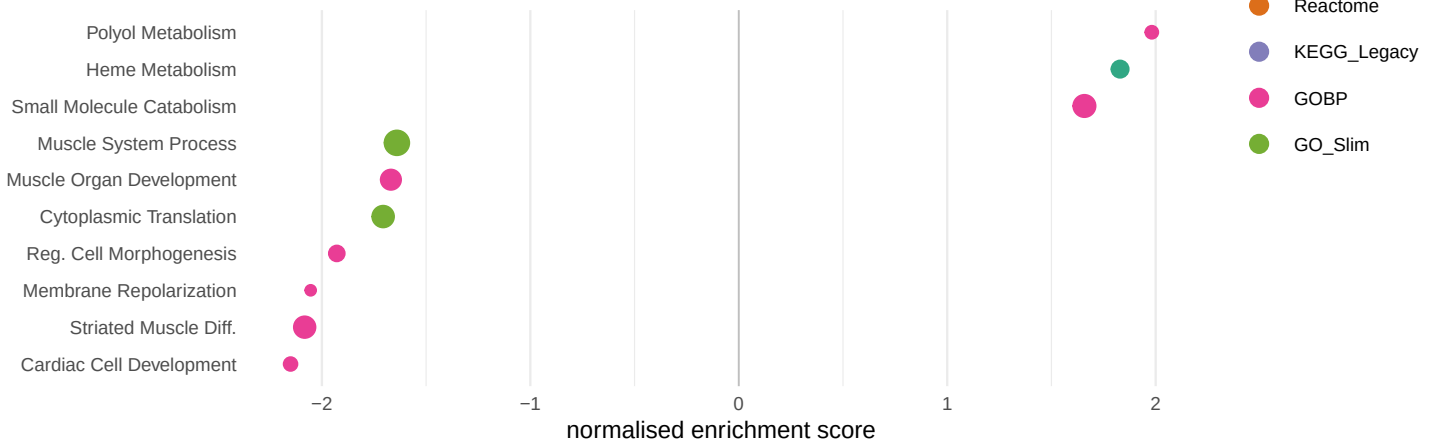
B

## HLRT\_Post-Pre, 10 of 68 survivors



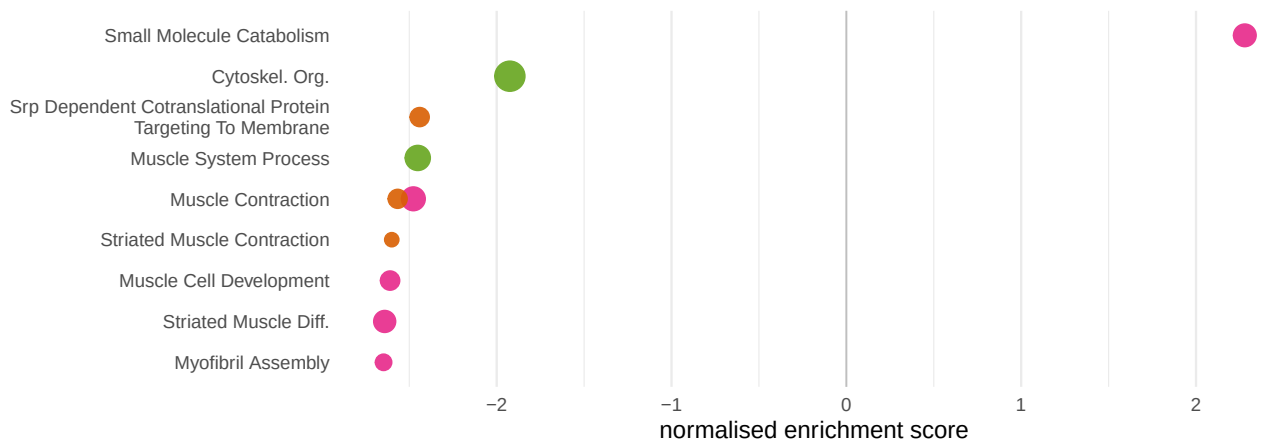
C

## Modality\_x\_Time\_Interaction, 10 of 10 survivors



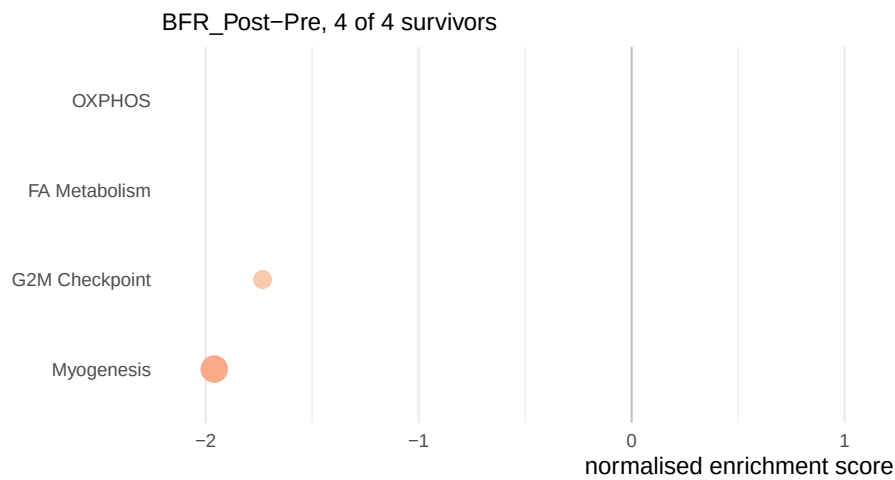
D

## BFR\_Post-HLRT\_Post, 10 of 52 survivors

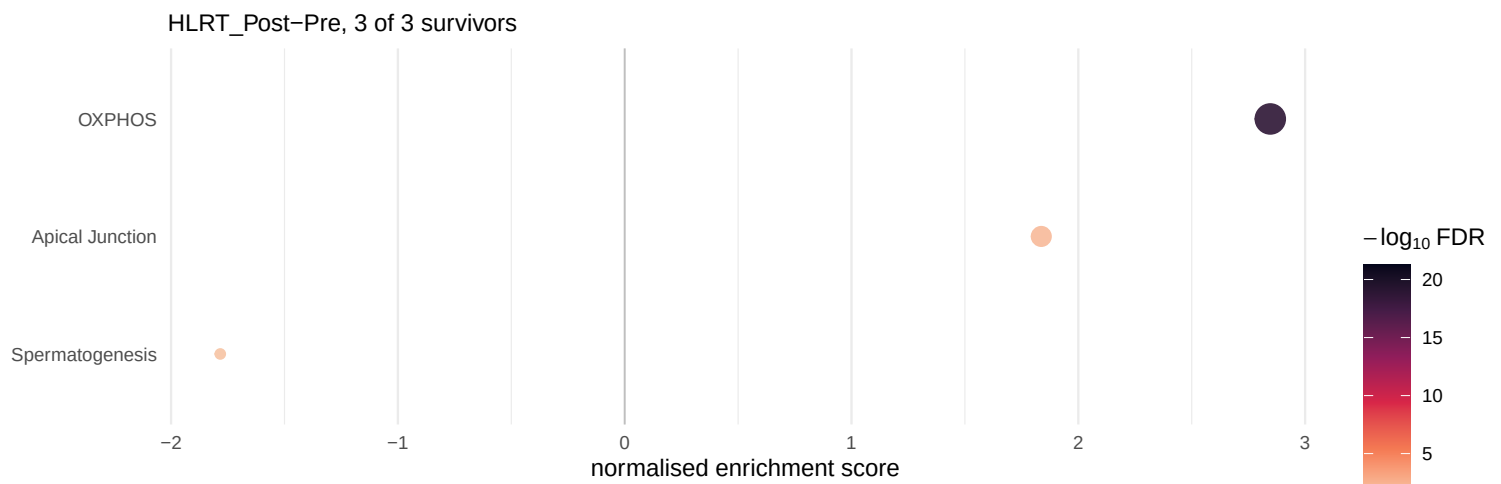


S1 Figure. Strongest fgsea sets after collapse, all collections. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) Modality\_x\_Time\_Interaction. (D) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is the collection. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.

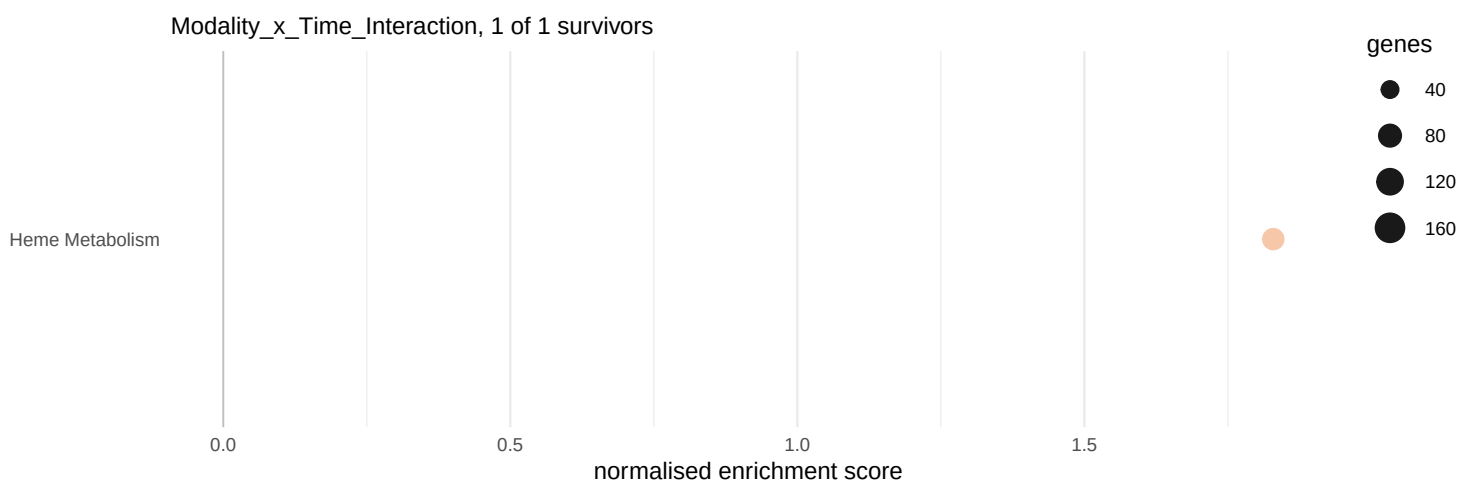
A



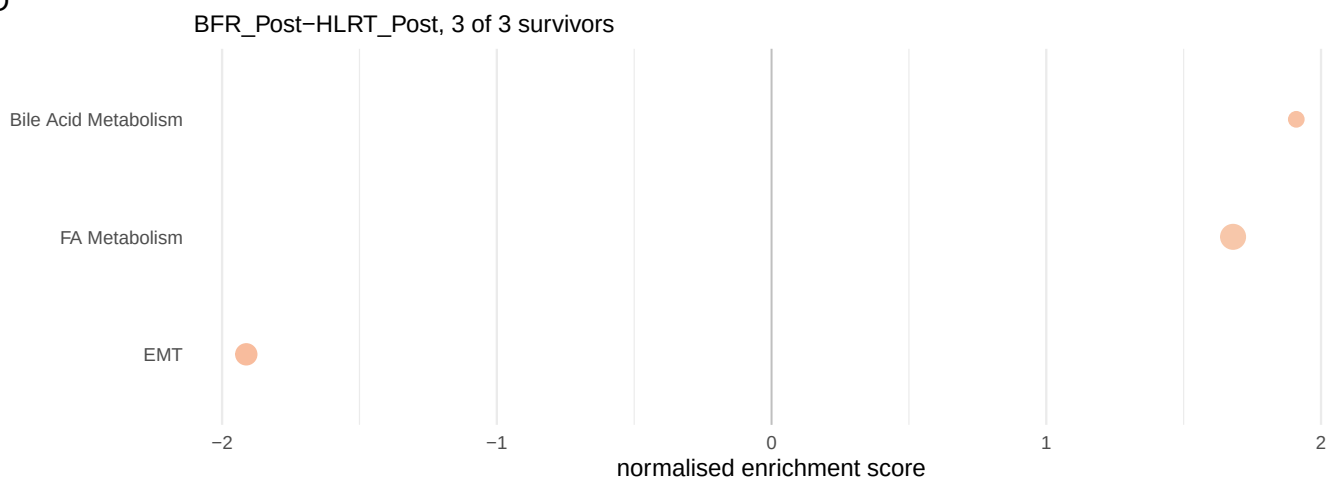
B



C

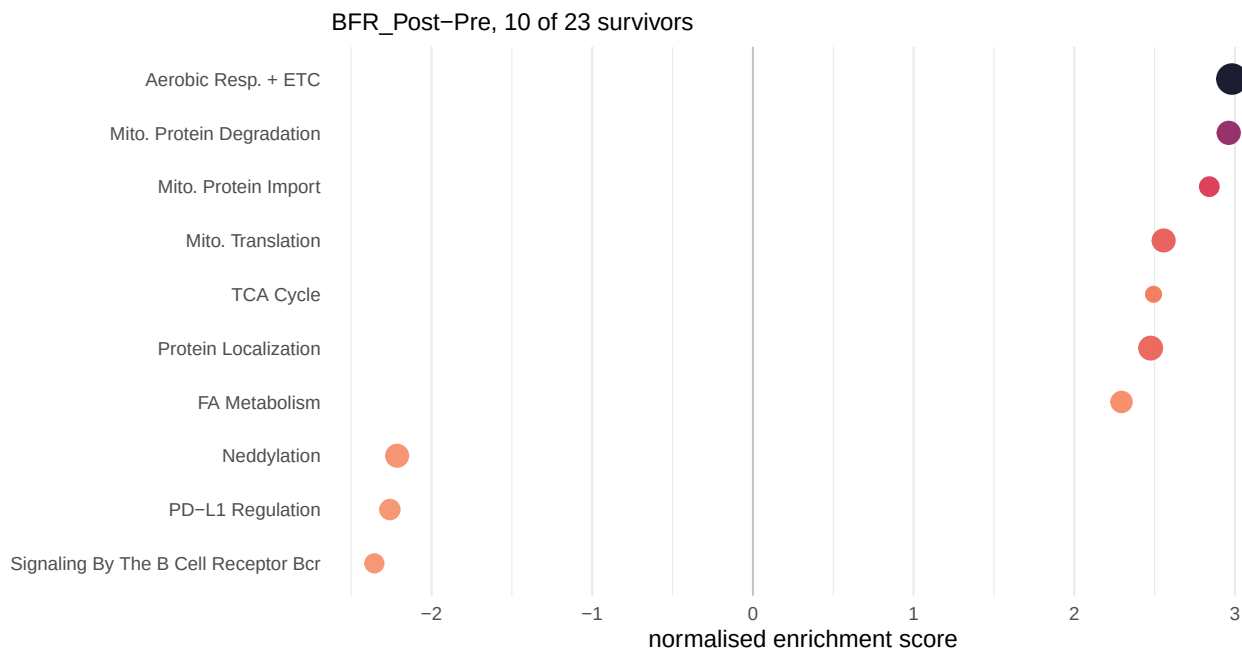


D

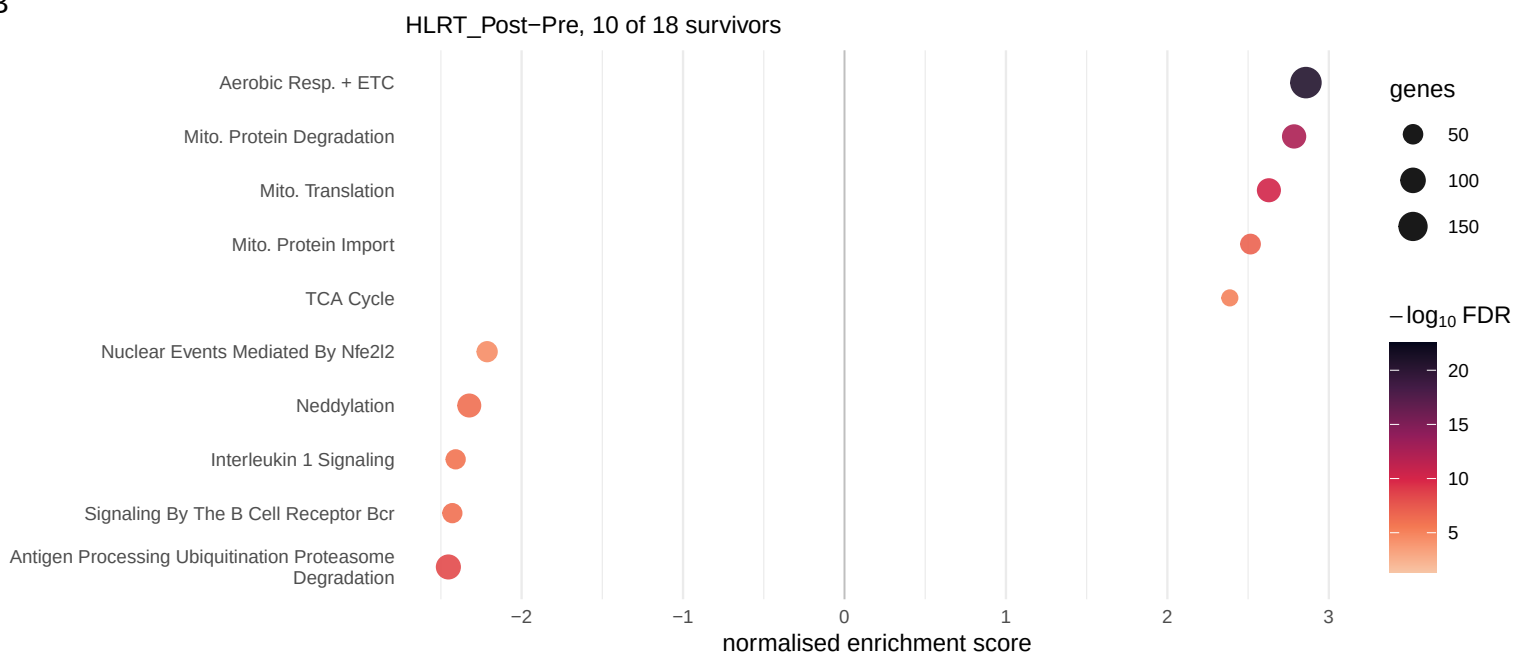


S2 Figure. Strongest fgsea sets after collapse, Hallmark. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) Modality\_x\_Time\_Interaction. (D) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is  $-\log_{10}$  FDR. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.

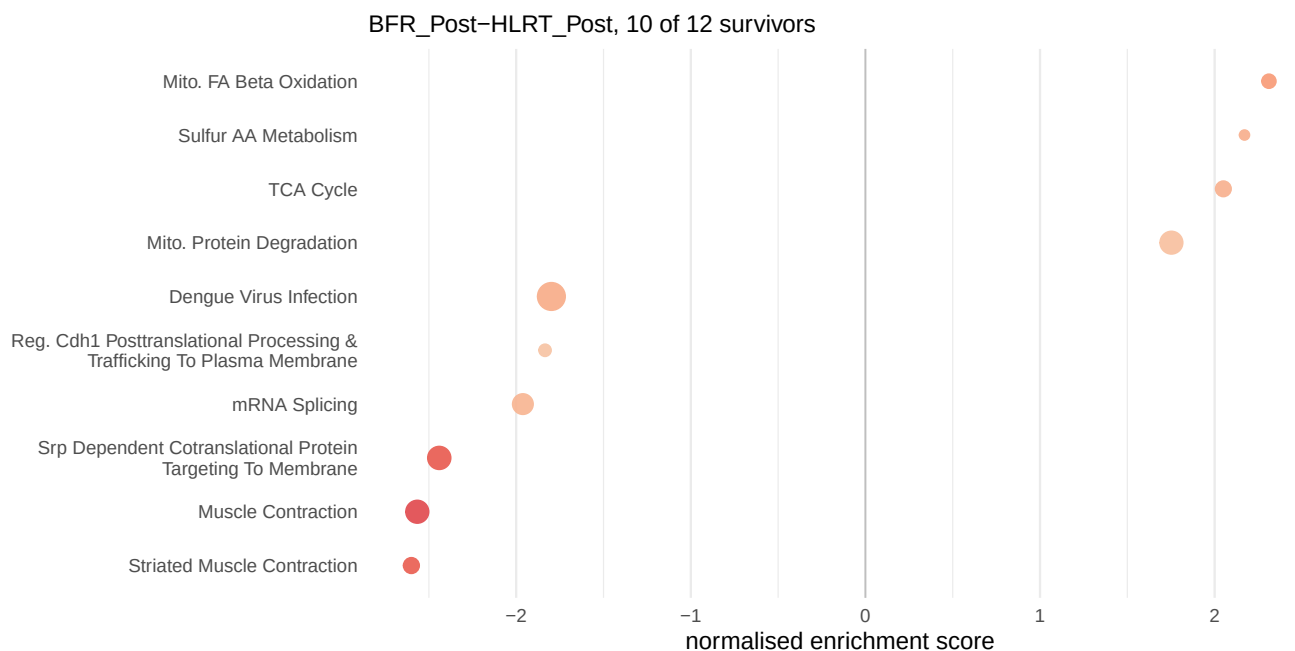
A



B

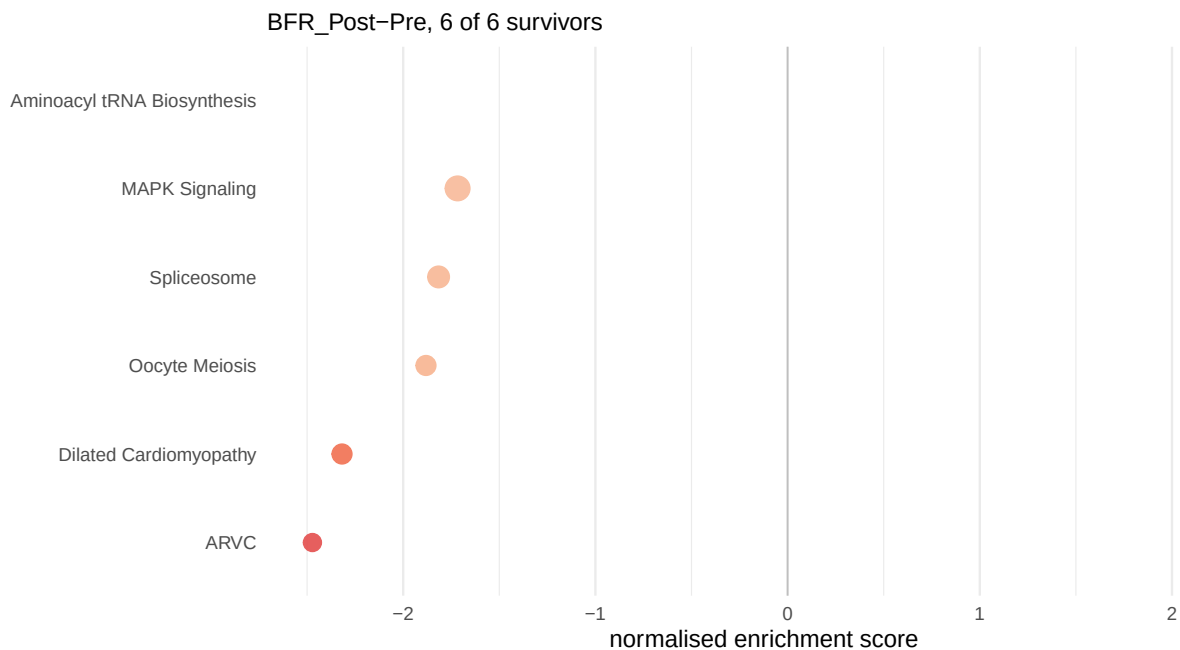


C

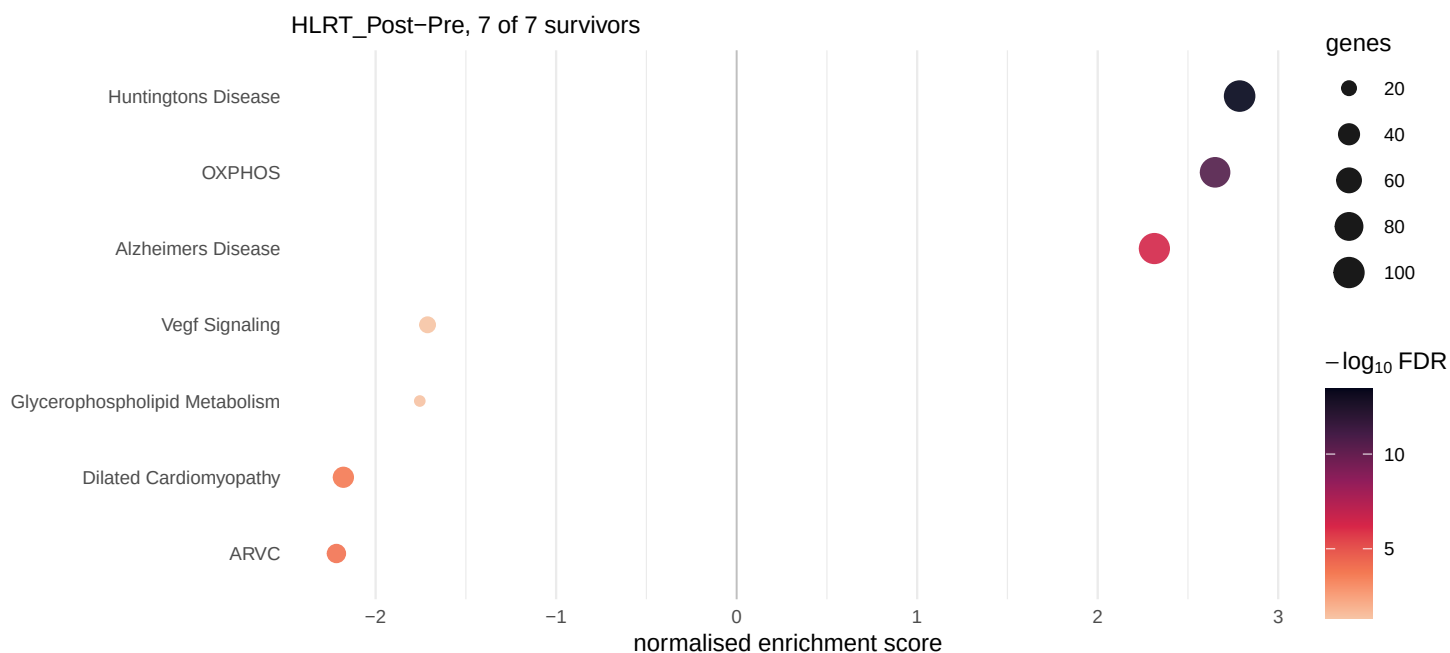


S3 Figure. Strongest fgsea sets after collapse, Reactome. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is  $-\log_{10}$  FDR. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.

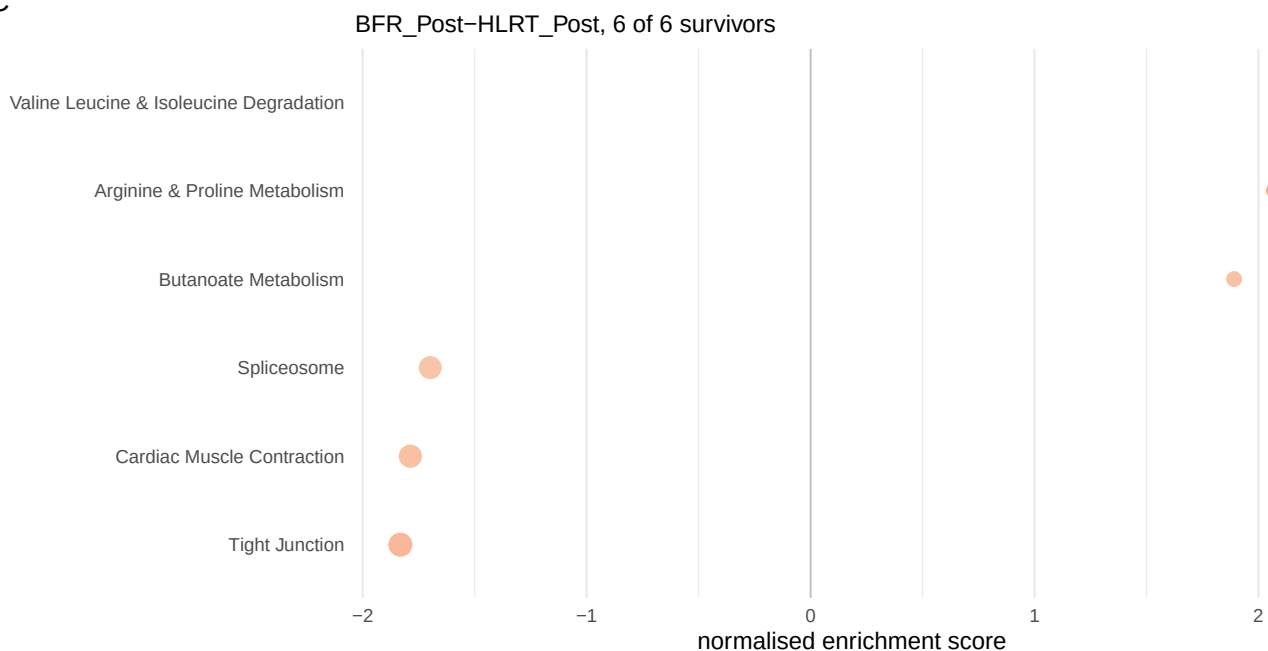
A



B



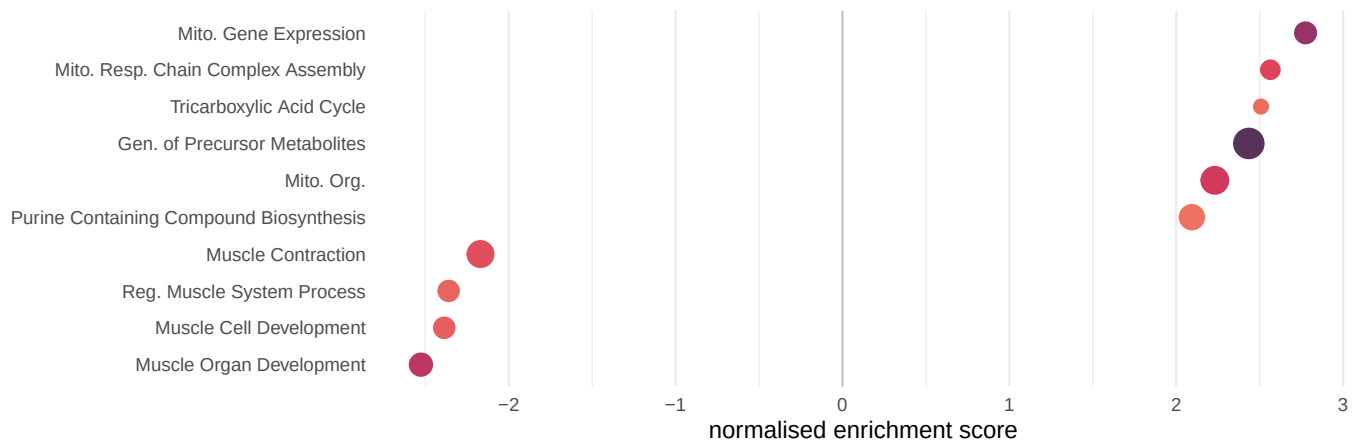
C



S4 Figure. Strongest fgsea sets after collapse, KEGG\_Legacy. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is  $-\log_{10}$  FDR. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.

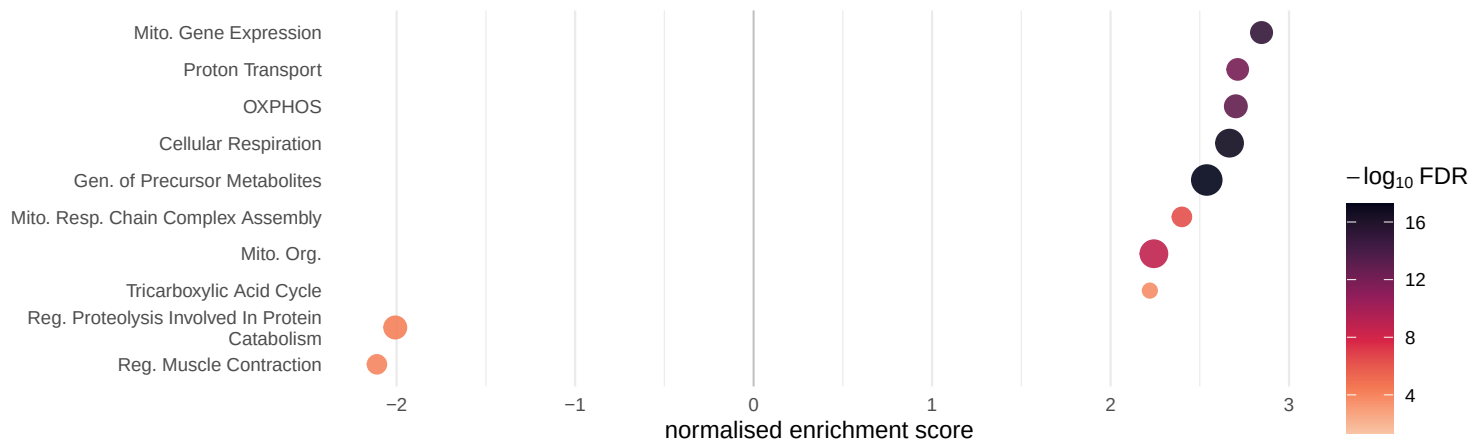
A

## BFR\_Post-Pre, 10 of 50 survivors



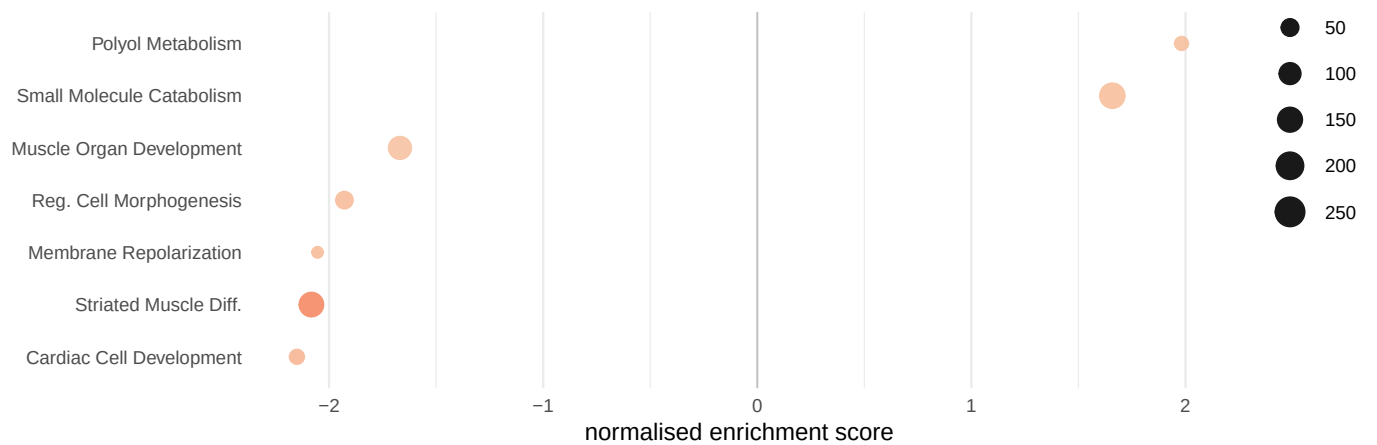
B

## HLRT\_Post-Pre, 10 of 39 survivors



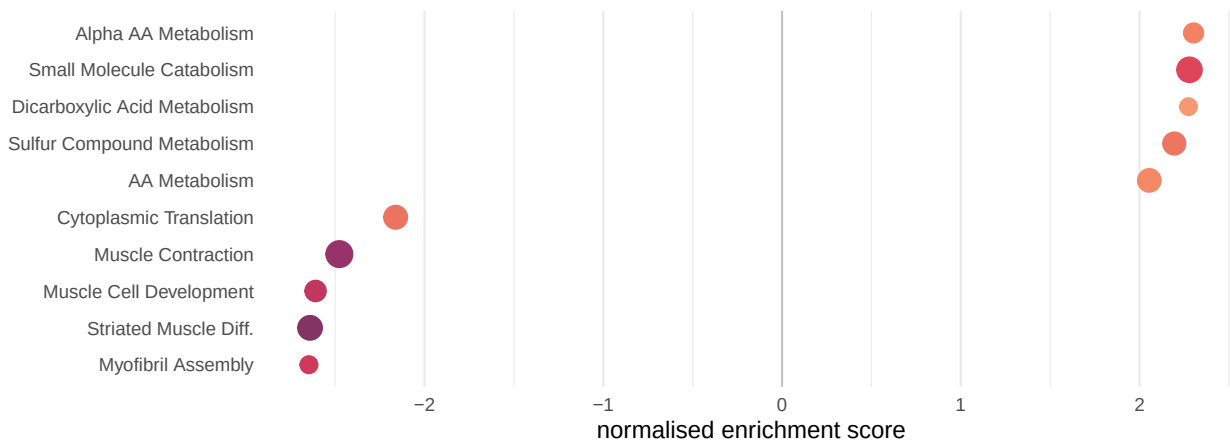
C

## Modality\_x\_Time\_Interaction, 7 of 7 survivors



D

## BFR\_Post-HLRT\_Post, 10 of 27 survivors



S5 Figure. Strongest fgsea sets after collapse, GOBP. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) Modality\_x\_Time\_Interaction. (D) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is  $-\log_{10}$  FDR. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.

A

BFR\_Post-Pre, 5 of 5 survivors

Nucleobase-Containing Small Molecule Metabolism

Protein Maturation

Nervous System Process

Mitotic Cell Cycle

DNA Repair

-2

-1

0

1

normalised enrichment score

B

HLRT\_Post-Pre, 1 of 1 survivors

DNA Repair

-1.5

-1.0

-0.5

0.0

normalised enrichment score

 $-\log_{10}$  FDR

9

6

3

C

Modality\_x\_Time\_Interaction, 2 of 2 survivors

Muscle System Process

Cytoplasmic Translation

-1.5

-1.0

-0.5

0.0

normalised enrichment score

genes

100

200

300

D

BFR\_Post-HLRT\_Post, 4 of 4 survivors

Lipid Metabolism

Cell Adhesion

Cytoskel. Org.

Muscle System Process

-2

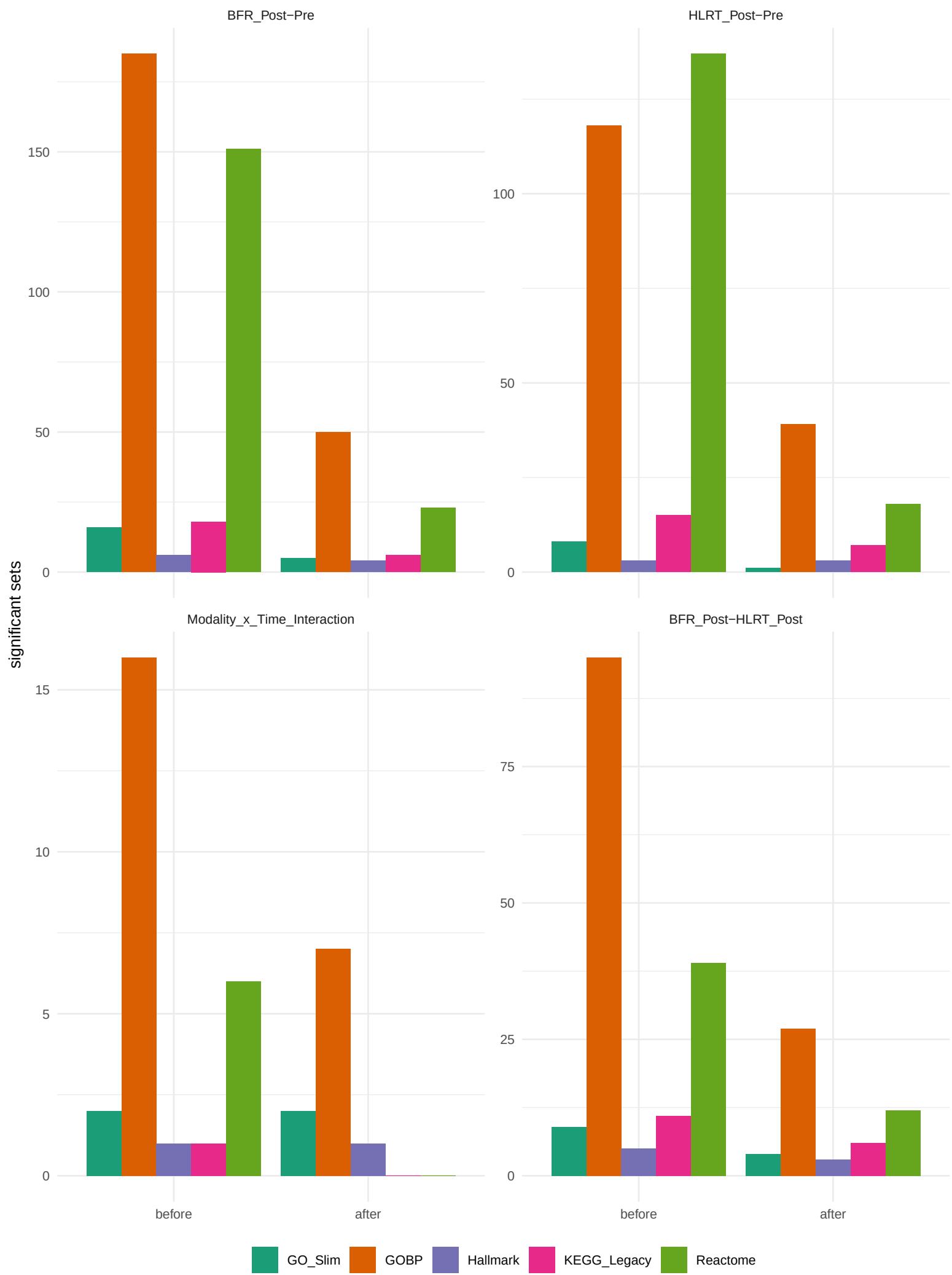
-1

0

1

normalised enrichment score

S6 Figure. Strongest fgsea sets after collapse, GO\_Slim. (A) BFR\_Post-Pre. (B) HLRT\_Post-Pre. (C) Modality\_x\_Time\_Interaction. (D) BFR\_Post-HLRT\_Post. Up to ten collapsePathways survivors per contrast, lowest adjusted p first; FDR is BH within contrast. Point size is the number of measured genes in the set; colour is  $-\log_{10}$  FDR. Contrasts with no survivor are omitted. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.



S7 Figure. Significant fgsea sets before and after collapse. Sets at FDR 0.05 per collection and contrast, before and after collapsePathways, which keeps a set only if it stands alone given a stronger set's leading edge. Data: set\_tests sheet of 01\_run\_fgsea\_and\_fry.xlsx.